BSc CS IT — PRACTICAL FILE

PRACTICAL / TUTORIAL NO. 12

AI-POWERED SOFTWARE TESTING

Application Under Test: SauceDemo (Swag Labs)

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Program	BSc CS IT

Includes workbook tables, test planning, defect log, and user-provided SauceDemo screenshots.

Test Cases

Tutorial 12 – AI-Powered Software Testing	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Website: SauceDemo (https://www.saucedemo.com/) | Use the demo site and capture your own page screenshots as practical evidence.							
TC ID	Requirement / Feature	Test Scenario	Test Data	Steps	Expected Result	Actual Result	Status
TC-01	Login	Login with valid credentials	standard_user / secret_sauce	1. Open website 2. Enter username 3. Enter password 4. Click Login	Products page opens successfully	To be filled during execution	Not Run
TC-02	Login	Login with incorrect password	standard_user / wrong123	1. Open website 2. Enter valid username 3. Enter incorrect password 4. Click Login	Error message is displayed; login is denied	To be filled during execution	Not Run
TC-03	Login	Login with empty fields	Username: blank; Password: blank	1. Open website 2. Leave fields empty 3. Click Login	Validation message is displayed	To be filled during execution	Not Run
TC-04	Products	Verify product list appears	Valid login	1. Log in 2. Observe Products page	Product names, prices, images and Add to cart buttons appear	To be filled during execution	Not Run
TC-05	Products	Add one item to cart	Sauce Labs Backpack	1. Log in 2. Click Add to cart for Backpack 3. Open cart	Selected product appears in cart and cart badge updates	To be filled during execution	Not Run
TC-06	Cart	Remove item from cart	One item already added	1. Open cart 2. Click Remove	Item is removed from cart	To be filled during execution	Not Run
TC-07	Sorting	Sort products by price low to high	Sort option: Price (low to high)	1. Log in 2. Select price low-to-high sort	Products display in ascending price order	To be filled during execution	Not Run
TC-08	Checkout	Continue checkout with valid details	First: Test; Last: User; ZIP: 395001	1. Add product 2. Open cart 3. Checkout 4. Enter details 5. Continue	Checkout overview displays item and price summary	To be filled during execution	Not Run
TC-09	Checkout	Submit checkout with missing postal code	First: Test; Last: User; ZIP: blank	1. Proceed to checkout 2. Leave postal code empty 3. Click Continue	Validation error is displayed	To be filled during execution	Not Run
TC-10	Checkout	Complete purchase	Valid checkout details	1. Reach checkout overview 2. Click Finish	Order confirmation page is displayed	To be filled during execution	Not Run
TC-11	Navigation	Logout from application	Logged-in user	1. Open menu 2. Click Logout	Login page is displayed	To be filled during execution	Not Run
TC-12	Session / Error	Try locked-out user	locked_out_user / secret_sauce	1. Enter locked-out user credentials 2. Click Login	Access is denied with an appropriate error message	To be filled during execution	Not Run

AI Test Data

AI-Generated Test Data	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Example test data for testing only. Do not enter real personal or payment informati on.							
Data ID	Purpose	Username	Password	First Name	Last Name	Postal Code	Expected Outcom e
TD-01	Valid login	standard_user	secret_sauce				Login succeeds
TD-02	Invalid password	standard_user	wrong123				Login rejected
TD-03	Unknown usernam e	unknown_user	secret_sauce				Login rejected
TD-04	Empty login						Validation error
TD-05	Valid checkout	standard_user	secret_sauce	Test	User	395001	Checkout can continue
TD-06	Missing postal code	standard_user	secret_sauce	Test	User		Validation error
TD-07	Locked-out account	locked_out_user	secret_sauce				Access denied
TD-08	Long input check	standard_user	secret_sauce	AAAAAAAAAAAA AAAAAAAAAAAA AAAAAA	BBBBBBBBBBB BBBBBBBBBBB BBBBBB	395001	Fields handle long values according to validation rules

Requirement Analysis

Requirement Analysis	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Map website functions to testable acceptance criteria.							
Req ID	Requirement	Acceptance Criteria	Risk	Suggested AI Technique	Related Test Cases	Priority	Notes
REQ-01	User authentication	Valid user can log in; invalid credentials are rejected	High	Generative AI test case generation	TC-01, TC-02, TC-03, TC-12	High	Check error messages and access control
REQ-02	Product listing	Products and prices are visible after login	Medium	Requirement summarization	TC-04	High	Check product names, prices and images
REQ-03	Cart operations	Items can be added to and removed from cart	High	Boundary and negative test generation	TC-05, TC-06	High	Check cart badge and item list
REQ-04	Product sorting	Products sort in selected order	Medium	Test scenario generation	TC-07	Medium	Verify ascending price sequence
REQ-05	Checkout validation	Required fields are validated before continuing	High	Negative test data generation	TC-08, TC-09	High	Test missing and valid fields
REQ-06	Order completion	Finish action shows order confirmation	High	Test script generation	TC-10	High	Verify confirmation page
REQ-07	Logout	User can end the session and return to login	Medium	Test flow generation	TC-11	Medium	Verify logged-out state

Defect Log

Defect Prediction & Defect Log	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Record only defects observed during actual execution; leave unobserved items blank.							
Defect ID	Related TC	Defect Title	Steps to Reproduce	Expected Result	Actual Result	Severity	Likelihood / Risk
BUG-01	TC-02	Invalid password validation	Enter standard_user and wrong password; click Login	Login denied with error	Fill after execution	Medium	Medium
BUG-02	TC-09	Missing postal code validation	Proceed to checkout and leave postal code blank	Validation error appears	Fill after execution	Medium	Medium

Selenium Plan

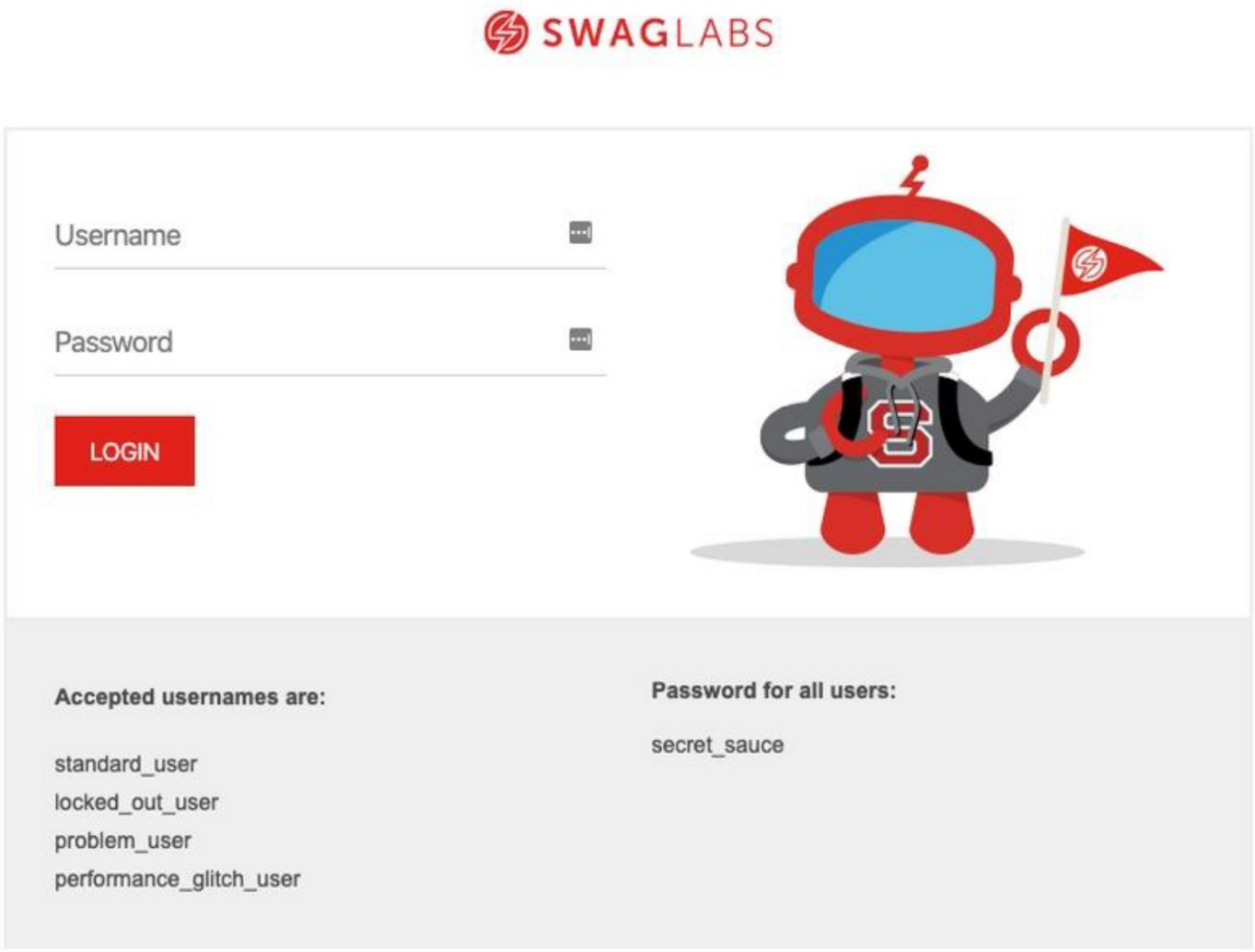
Selenium Automation Plan	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Use this as a planning sheet for generated Selenium scripts; verify selectors and behavior on the live demo site.							
Script ID	Test Objective	Page / Flow	Automation Steps	Assertions	Suggested Locator Strategy	Execution Result	Notes
SEL-01	Valid login	Login → Products	Open URL; locate username and password fields; enter credentials; click Login	Products page title or inventory container is visible	Prefer stable IDs such as user-name, password, login-button	Not Run	Confirm locators in browser
SEL-02	Invalid login	Login	Enter valid username and wrong password; click Login	Error container is visible	Use stable error element selector	Not Run	Avoid relying only on text if localization changes
SEL-03	Add item to cart	Products → Cart	Log in; click Add to cart for a product; open cart	Selected product appears in cart	Use stable product button IDs where available	Not Run	Verify cart badge count
SEL-04	Checkout validation	Cart → Checkout	Add item; checkout; enter first and last name; omit postal code; continue	Validation error is displayed	Use stable field IDs and error element	Not Run	Do not use real personal information
SEL-05	Complete purchase	Checkout → Confirmation	Add item; checkout with test data; continue; finish	Order confirmation message appears	Use stable buttons and confirmation container	Not Run	Keep test independent and repeatable

Screenshot Checklist

Website Screenshot Checklist	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5
Capture screenshots of the actual SauceDemo website pages (not code screenshots) and insert/paste them into your practical report.					
No.	Page / Screen	How to Capture	Suggested File Name	Captured? (Yes/No)	Observation
1	Login Page	Open https://www.saucedemo.com/ before entering credentials	01_Login_Page.png	No	Show username and password fields
2	Products Page	Log in using standard_user / secret_sauce	02_Products_Page.png	No	Show product list and prices
3	Cart Page	Add an item and open the cart	03_Cart_Page.png	No	Show selected product
4	Checkout Information	Click Checkout from cart	04_Checkout_Info.png	No	Show first name, last name and postal code fields
5	Checkout Overview	Enter test details and click Continue	05_Checkout_Overview.png	No	Show item summary and total
6	Order Confirmation	Click Finish	06_Order_Confirmation.png	No	Show order completion message
7	Login Error	Try invalid password and capture the error	07_Login_Error.png	No	Show error message

SCREENSHOT EVIDENCE

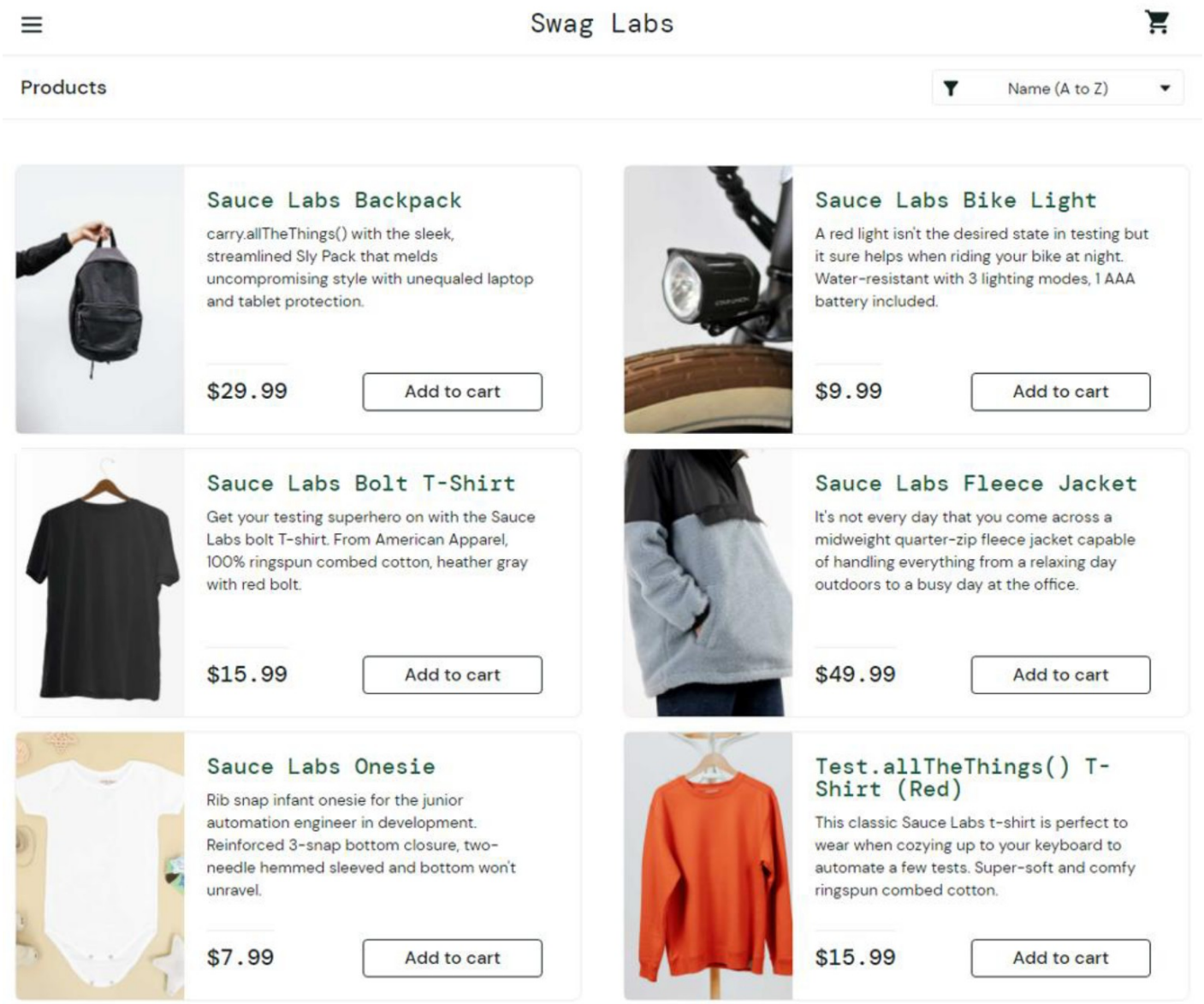
Screenshot 1 — SauceDemo Login Page



Note: Screenshots supplied by the student are included as practical evidence/reference.

SCREENSHOT EVIDENCE

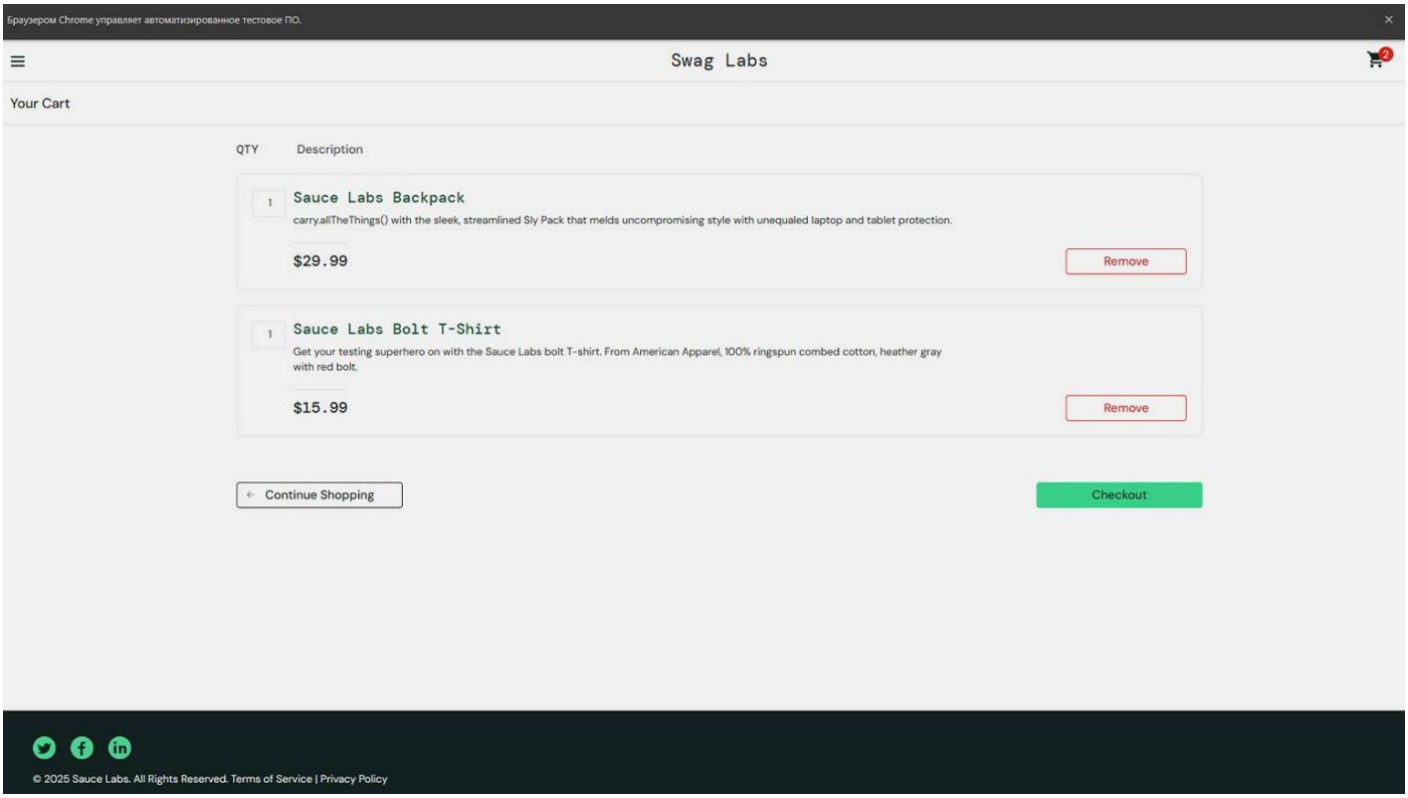
Screenshot 2 — Products Page / Product Listing



Note: Screenshots supplied by the student are included as practical evidence/reference.

SCREENSHOT EVIDENCE

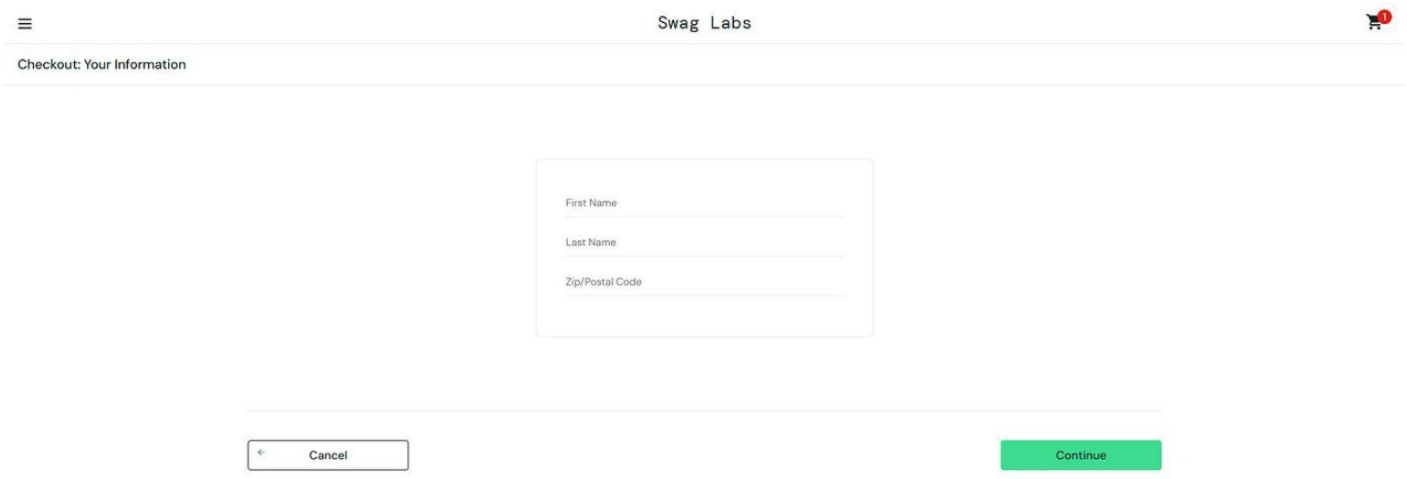
Screenshot 3 — Shopping Cart with Selected Products



Note: Screenshots supplied by the student are included as practical evidence/reference.

SCREENSHOT EVIDENCE

Screenshot 4 — Checkout: Your Information



Note: Screenshots supplied by the student are included as practical evidence/reference.